

# Hallucination Firewall

*Trust AI Before You Trust Its Answer.*

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## Real-Time Verification Middleware for Enterprise AI Outputs

Business Plan & Startup Blueprint

Prepared for: Idea Evaluation & Comparative Business Modeling Assignment

Category: Enterprise AI Infrastructure / Trust & Safety / RegTech

Target Sectors: LegalTech, Insurance, Healthcare, FinTech, Government

Confidential — Prepared as part of a structured startup idea validation exercise. Figures are estimates for planning purposes and should be revalidated with primary research before investor submission.

# 1. Vision & Mission

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## Vision

Become the trust infrastructure for Enterprise AI by verifying every AI-generated statement before it reaches a human.

## Mission

Enable enterprises to deploy AI safely in high-risk workflows by automatically detecting hallucinations, unsupported claims, missing citations, and contradictions.

## Positioning

Hallucination Firewall is positioned to become the “Cloudflare for AI” — just as every web application needs cybersecurity, every AI application will need an AI firewall sitting between the model and the human who acts on its output.

# 2. Problem Statement

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## 2.1 Current Situation

Large Language Models frequently generate hallucinated facts, fabricated citations, wrong legal references, incorrect financial calculations, unsupported medical recommendations, and false compliance statements. Because these outputs read as confident and fluent, humans downstream often accept them without independent verification.

## 2.2 Consequences

- Legal liability from acting on fabricated or incorrect information.
- Direct financial losses from decisions made on false AI outputs.
- Regulatory penalties for non-compliant AI-generated statements.
- Loss of customer trust once errors surface publicly.
- Outright compliance failures in regulated industries.

## 2.3 Market Pain, By Sector

| Sector                | Failure Chain                                                                  |
|-----------------------|--------------------------------------------------------------------------------|
| Legal Firms           | Wrong case citation → wrong legal advice → professional liability              |
| Insurance             | AI approves invalid claims → direct financial loss                             |
| Healthcare            | Incorrect discharge summary → patient risk                                     |
| FinTech               | Wrong RBI regulation quoted → compliance issue                                 |
| Enterprises (general) | Employees blindly trust AI → business decisions based on incorrect information |

# 3. Customer Personas

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| Persona                      | Who / Pain                                                                               |
|------------------------------|------------------------------------------------------------------------------------------|
| Primary: Enterprise AI Teams | CTO, Head of AI, AI Engineering Lead, AI Governance Team — “We cannot trust AI outputs.” |
| Secondary: Legal Firms       | Need every legal statement verified before it goes to a client.                          |
| Third: Insurance Companies   | Need AI-generated claims decisions verified before approval.                             |
| Fourth: Healthcare Providers | Need medical documentation grounded in actual patient records.                           |
| Fifth: Banks & FinTech       | Need regulatory compliance validation on every AI-generated statement.                   |

## 4. Jobs To Be Done

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Across every persona, customers need to:

- Verify AI outputs before anyone acts on them.
- Detect hallucinations automatically rather than relying on manual spot-checks.
- Ensure citations actually exist and actually support the statement made.
- Prevent compliance violations before they reach a regulator or a customer.
- Reduce the amount of manual review currently required on every AI output.
- Build organizational trust in AI so it can be deployed in higher-stakes workflows.

## 5. Existing Alternatives & Their Weaknesses

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| Existing Approach           | Weakness                                         |
|-----------------------------|--------------------------------------------------|
| Manual Review               | Slow                                             |
| Human QA                    | Expensive                                        |
| GPT Self-Check              | Not reliable                                     |
| Basic RAG                   | Doesn't verify reasoning, only retrieves context |
| Prompt Engineering          | Doesn't eliminate hallucination                  |
| Guardrails (existing tools) | Mostly rule-based, not true verification         |

### The Opportunity

AI adoption inside enterprises is expanding rapidly, and every enterprise AI system that touches a real decision will eventually need a verification layer sitting between the model and the human — the same way every web application eventually needed a cybersecurity layer.

## 6. Product Overview

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Hallucination Firewall is a middleware layer that sits between the LLM and the human recipient of its output, checking every statement before it is trusted or acted upon.

### 6.1 Product Workflow

Enterprise Data → RAG → LLM → Hallucination Firewall → Verification Engine → Confidence Score → Human Review.

## 7. Core Features

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| Feature                    | What It Does                                                                                                                                                                                                      |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Claim Extraction        | Breaks an AI response into atomic, checkable claims. Example: “RBI issued circular ABC in 2024” is decomposed into the entity (RBI), the object (Circular ABC), and the date (2024) for independent verification. |
| 2. Source Verification     | Searches across PDFs, the enterprise knowledge base, policies, contracts, medical records, and regulations to confirm each extracted claim against a real source.                                                 |
| 3. Citation Validation     | Checks both that a cited source actually exists and that it actually supports the statement being made — not just that a citation is present.                                                                     |
| 4. Contradiction Detection | Flags cases where the source material says one thing (A) and the AI output says another (B).                                                                                                                      |
| 5. Confidence Scoring      | Assigns every statement a rating of High Confidence, Medium Confidence, Low Confidence, or Unsupported.                                                                                                           |
| 6. Explainability          | Shows why a statement is or is not supported, displaying the relevant paragraph, highlighted evidence, and the exact document reference.                                                                          |
| 7. Risk Classification     | Labels every statement as Safe, Needs Review, Hallucinated, Missing Source, or Contradicted.                                                                                                                      |
| 8. Enterprise Dashboard    | Surfaces hallucination rate, confidence trend over time, most frequent error types, user acceptance rate, and exportable compliance reports.                                                                      |

## 8. AI Pipeline & Product Architecture

### 8.1 End-to-End AI Pipeline

User Query → RAG → LLM → Claim Extractor → Retriever → Evidence Matcher → Fact Checker → Confidence Scorer → Risk Engine → Verified Response.

### 8.2 Product Architecture

LLM → Firewall → Claim Engine → Retriever → (Evidence Database and Scoring Engine in parallel) → Policy Engine → Verified Output.

### 8.3 Core AI Components

- **Claim Extraction Engine:** converts paragraphs into discrete factual statements.
- **Retrieval Engine:** retrieves the top-K supporting documents for each claim.
- **Semantic Similarity Engine:** measures how closely a statement matches its evidence.
- **Rule Engine:** checks dates, names, numbers, regulations, and cross-references deterministically.
- **Citation Engine:** ensures every statement carries verifiable evidence.
- **Hallucination Detector:** classifies each claim as Unsupported, Contradicted, Weak Evidence, or Strong Evidence.

### 8.4 Product Modules

- Core API — REST API and SDK for direct integration into existing AI pipelines.
- Enterprise Dashboard — analytics on verification outcomes across the organization.
- Audit Logs — complete, exportable verification history for every response.
- Compliance Reports — exportable in PDF and CSV for regulators and auditors.
- Admin Portal — policy configuration, user management, and access control.

## 9. Technology Stack

| Layer                | Stack                                  |
|----------------------|----------------------------------------|
| Frontend             | React, Next.js                         |
| Backend              | FastAPI, Python                        |
| Vector Database      | Pinecone, Weaviate, Milvus             |
| Database             | PostgreSQL                             |
| AI / LLM Providers   | OpenAI, Gemini, Claude, Llama, Mistral |
| Retrieval Frameworks | LangChain, LlamaIndex                  |
| Infrastructure       | Kubernetes, Docker, AWS, Azure, GCP    |

## 10. Target Industries & Competitive Landscape

### 10.1 Phased Industry Rollout

| Phase   | Industries                                                            |
|---------|-----------------------------------------------------------------------|
| Phase 1 | LegalTech, Insurance, Healthcare, FinTech                             |
| Phase 2 | Government, Defense Contractors, Life Sciences, Pharma, Manufacturing |
| Phase 3 | General Enterprise AI (horizontal expansion)                          |

### 10.2 Competitive Landscape

| Company       | Gap Relative to Hallucination Firewall                          |
|---------------|-----------------------------------------------------------------|
| Giskard       | Focused on ML testing, not live production verification         |
| Patronus AI   | US-centric, limited India regulatory alignment                  |
| TruLens       | Developer tooling, not an enterprise compliance product         |
| LangSmith     | Observability only — does not verify claims against evidence    |
| Guardrails AI | Primarily rule-based rather than evidence-grounded verification |
| Galileo AI    | Positioned as an evaluation platform, not a real-time firewall  |

### 10.3 Competitive Advantage

- Enterprise middleware that sits in the live request path, not just an offline eval tool.
- Real-time verification rather than batch or after-the-fact review.
- Works with every major LLM provider rather than locking into one model.
- Domain-specific verification tuned per industry (legal, medical, financial, regulatory).
- Compliance-first architecture built around audit trails and exportable reports.
- India-first regulatory support (RBI, IRDAI, DPDP) alongside global standards.
- API-first platform designed for straightforward integration into existing AI pipelines.

## 11. Revenue Model & Pricing

### 11.1 SaaS Subscription Tiers

| Tier         | Price               |
|--------------|---------------------|
| Starter      | Rs. 25,000 / month  |
| Professional | Rs. 75,000 / month  |
| Enterprise   | Rs. 3 Lakh+ / month |

### 11.2 Other Revenue Lines

- API pricing charged per 1,000 verified responses processed.
- Enterprise licensing via annual contracts for large accounts.
- Professional services covering deployment, customization, integration, and training.

### **11.3 Pricing Strategy / Funnel**

Pilot (free) → Proof of Concept (Rs. 2–5 Lakhs) → Annual Enterprise Contract (Rs. 25 Lakhs – Rs. 2 Crore).



## 12. Go-To-Market Strategy

### 12.1 Ideal Customer Profile

Companies that are already deploying AI in production, with 50+ employees.

### 12.2 Sales Motion

Founder-led sales → Enterprise pilots → Proof of Concept → Annual contracts.

### 12.3 Acquisition Channels

- AI conferences
- LegalTech events
- CIO roundtables
- LinkedIn outbound and content
- AI consulting partners
- Microsoft Marketplace listing
- AWS Marketplace listing

## 13. Unit Economics

| Metric                              | Value         |
|-------------------------------------|---------------|
| Customer Acquisition Cost (CAC)     | Rs. 3–8 Lakhs |
| Average Annual Contract Value (ACV) | Rs. 40 Lakhs  |
| Gross Margin                        | 85%           |
| Sales Cycle                         | 3–6 months    |
| Lifetime Value (LTV)                | Rs. 2 Crore+  |

An ACV of Rs. 40 Lakhs against a CAC of Rs. 3–8 Lakhs, combined with an 85% gross margin, implies a strong LTV:CAC ratio once a customer converts — the primary risk to manage is the 3–6 month enterprise sales cycle, not unit profitability once landed.

## 14. 120-Day Execution Plan

| Month   | Focus                                                                                       | Deliverable       |
|---------|---------------------------------------------------------------------------------------------|-------------------|
| Month 1 | Customer discovery — interview banks, hospitals, insurance companies, and law firms         | Validated Problem |
| Month 2 | Build prototype — claim extraction, evidence search, confidence scoring                     | Technical MVP     |
| Month 3 | Pilot with 5 enterprise customers; measure detection accuracy, false positives, and latency | Pilot Results     |

|         |                                                                                                 |                    |
|---------|-------------------------------------------------------------------------------------------------|--------------------|
| Month 4 | Enterprise-ready hardening — dashboard, APIs, RBAC, audit logs, security, deployment automation | Commercial Product |
|---------|-------------------------------------------------------------------------------------------------|--------------------|

## 15. Success Metrics

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### 15.1 Product KPIs

- Verification accuracy
- Hallucination detection rate
- Precision
- Recall
- Latency
- False positive rate

### 15.2 Business KPIs

- ARR and MRR
- Average Contract Value (ACV)
- Gross margin
- CAC
- LTV
- Number of enterprise customers

### 15.3 Customer KPIs

- AI Trust Score
- Review time saved
- Compliance incidents prevented
- Hallucination reduction achieved

## 16. Compliance

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- DPDP (India) readiness
- ISO 27001
- SOC 2
- HIPAA for healthcare deployments
- GDPR
- RBI guidelines for FinTech customers
- IRDAI guidelines for insurance customers
- Full audit trail on every verification
- Role-based access control

## 17. Risks

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| Category   | Risk                                                                               |
|------------|------------------------------------------------------------------------------------|
| Technical  | Retrieval quality, multi-document verification complexity, latency overhead        |
| Business   | Long enterprise sales cycle; integration complexity with existing AI stacks        |
| Regulatory | Shifting AI governance requirements; data privacy obligations across jurisdictions |

## 18. Funding Strategy

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| Stage    | Sources                                                                       |
|----------|-------------------------------------------------------------------------------|
| Pre-Seed | Founder capital; incubators; government grants                                |
| Seed     | Enterprise SaaS investors; AI-focused VCs; strategic angels                   |
| Series A | Global enterprise AI funds; cybersecurity investors; infrastructure investors |

## 19. Long-Term Vision

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Become the universal trust layer for AI: every AI response generated anywhere should pass through the Hallucination Firewall before reaching a human.

### Future Product Roadmap

- AI Agent Monitoring
- AI Risk Scoring
- Multi-Agent Verification
- AI Compliance Copilot
- Continuous AI Auditing
- AI Governance Platform
- Enterprise AI Observatory
- Real-time Policy Engine
- Autonomous Risk Management

### NORTH STAR METRIC

**Verified AI Responses Processed Per Day**

*Every AI answer. Verified. Explained. Trusted.*